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Bayesian global optimization : Bgolearn Framework & Implement

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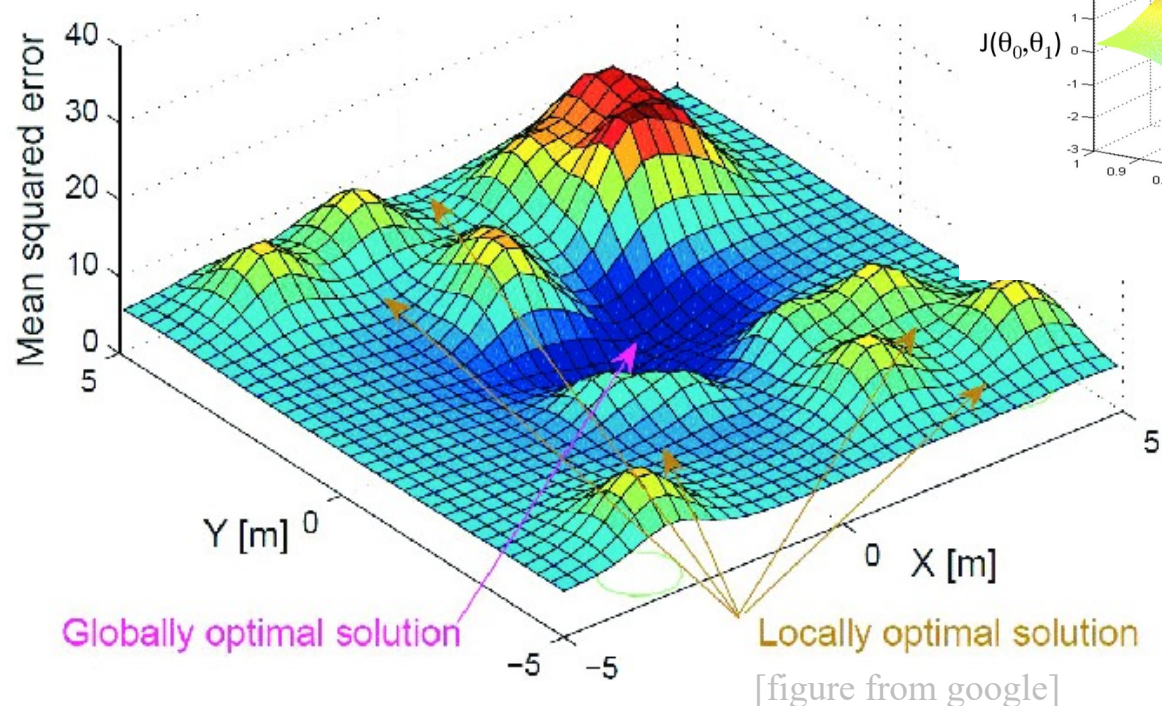
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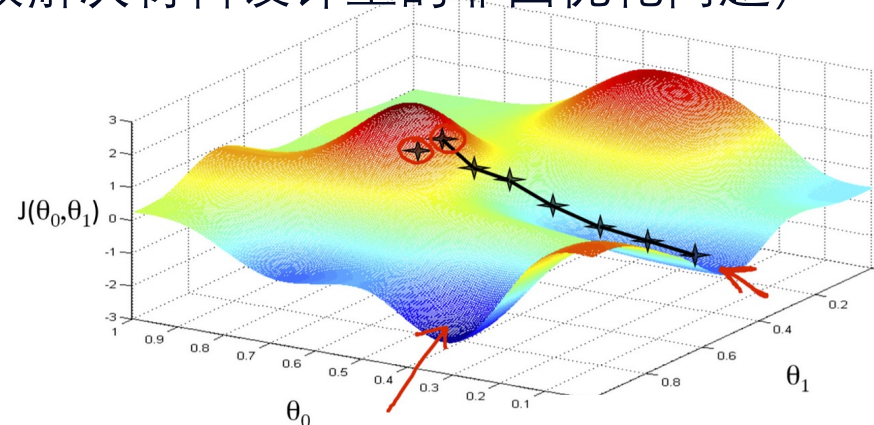


Non-convex Optimization 基础概念

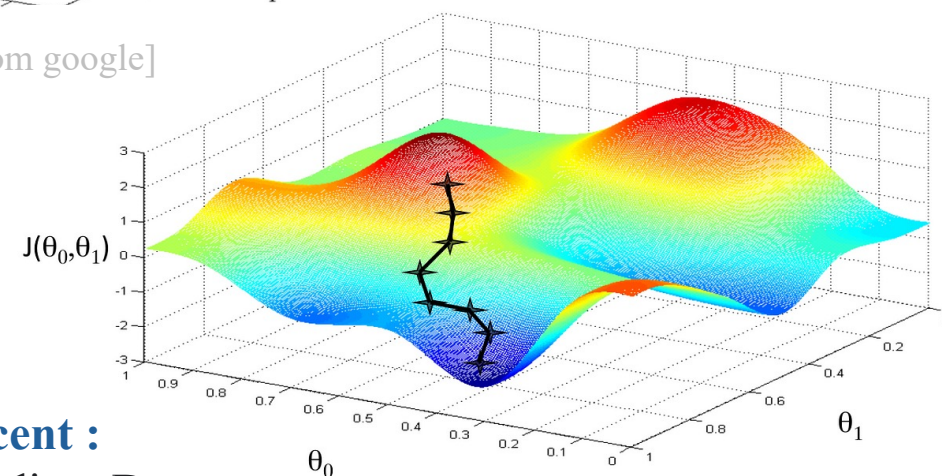
A **non-convex optimization** problem is characterized by having a non-convex objective function or non-convex constraints. (Bgolearn 旨在高效解决材料设计里的非凸优化问题)



The function is **expressed analytically**



[figure from google]



[figure from google]

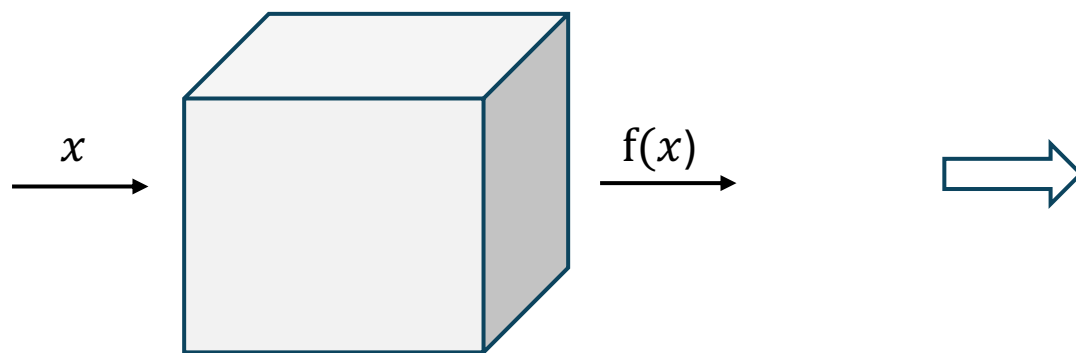
Gradient descent :
Stochastic Gradient Descent
Steepest Gradient Descent
Newton Gradient Descent ...



Non-convex Optimization 问题定义

Goal : solve $\operatorname{argmin}_x f(x)$

Where $f(x)$ is a black box and is a **time-consuming-to-evaluate** function



Difficulties :

The analytical expression of the function is **not known**

The computation of $f(x)$ is associated with **significant time cost**

Heuristic optimization algorithm:

Particle swarm algorithm

Simulated annealing algorithm

Genetic algorithm

...

Bayesian optimization:

Expected improvement algorithm

Upper confidence bound

Predictive entropy search

Knowledge gradient

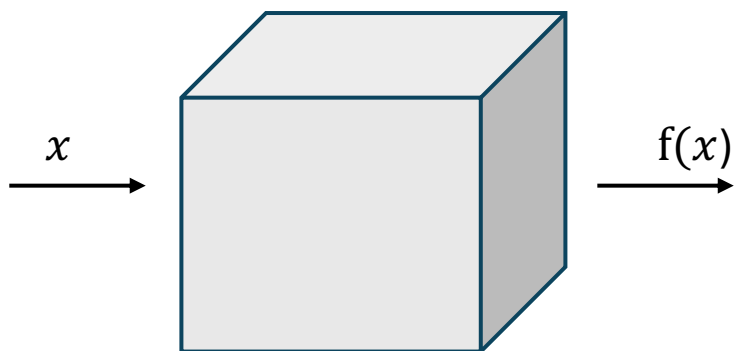
...

梯度不能作为有效信息时：启发式or贝叶斯



Target of Optimization (有效采样问题)

The optimization algorithms applied to black-box problems are collectively geared towards solving the challenge of effective sampling



$$x_1 \rightarrow f(x_1)$$

$$x_2 \rightarrow f(x_2)$$

...

$$x_n \rightarrow f(x_n)$$

(n is a **finite number**)

$$x_* \rightarrow f(x_*)$$

Many optimization algorithms leverage historical information to guide the selection of the next sampling point, determining which infill data,

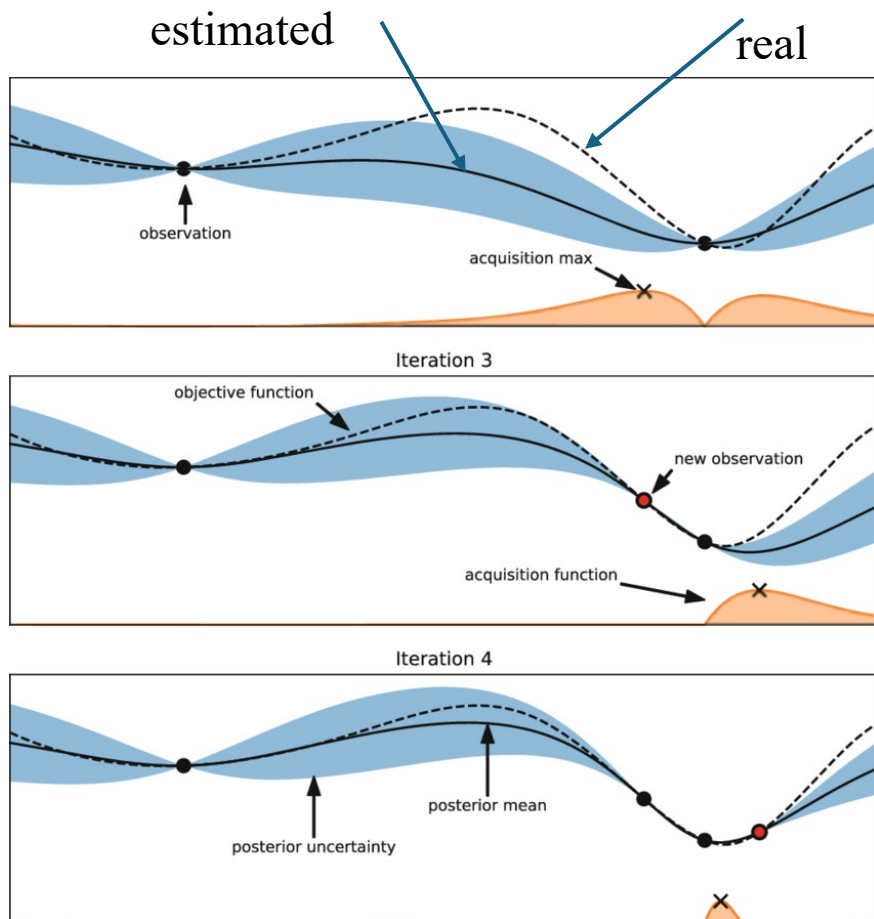
$x_{n+1} \rightarrow f(x_{n+1})$ is most informative in the pursuit of deriving the optimal solution x_* .

如何有效采样？什么是有效的采样？



The Sampling (主动学习过程)

All optimization algorithms incorporate a fundamental concept of an **optimization (response) surface**



[figure from google]

The optimization surface **undergoes changes** with the introduction of infill data.

This response surface can be **implicit**, as seen in heuristic optimization, or **explicit**, as **observed** in Bayesian optimization.

Bayesian optimization strategy **does not unconditionally trust the derived response surface**. Instead, it carefully navigates the **exploration-exploitation trade-off**, balancing the need for exploring new regions with exploiting the current knowledge to achieve optimal results.



Material Design (材料设计)

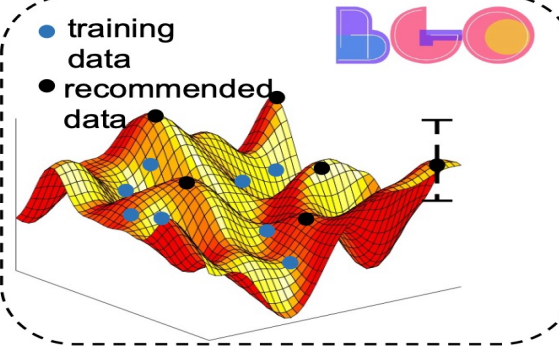
History informations

datasets

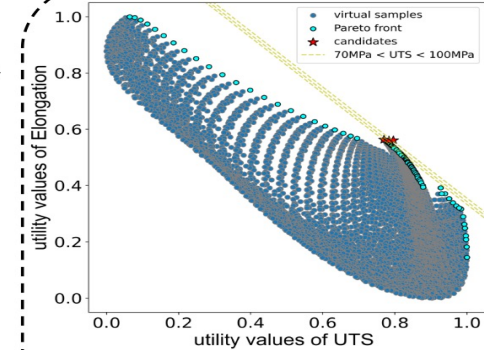
Sn	Ag	Cu	Bi	In	Ti	UTS	EL
Bal.	1	0.5	3	2.5	0.1	64.60	28.85
Bal.	1	0.5	4	2.5	0.1	77.65	16.44
Bal.	1	0.5	5	3.5	0.3	89.85	11.91
⋮							

Response surface

gaussian proc.



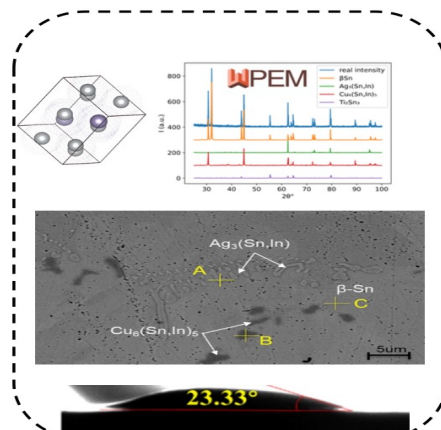
optimization



satisfied ?

Yes

No



characterization

Sn	Ag	Cu	Bi	In	Ti	UTS	EL
Bal.	1	0.5	5.5	5.5	0.7	106.7	11.8
Bal.	1	0.5	2.5	3.3	0.1	62.7	26.4
Bal.	1	0.5	1.5	4.4	0.2	73.9	24.4

experimental validation



New infill data

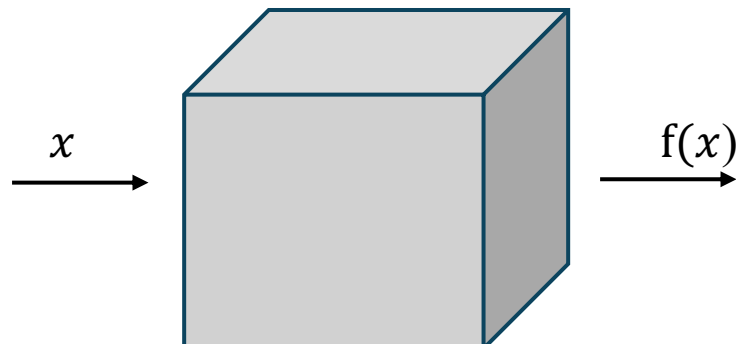
Multi-targets balance
& candidate evaluation

Cao Bin et al, Active learning accelerates the discovery of high strength and high ductility lead-free solder alloys. Materials & Design 241, 112921.



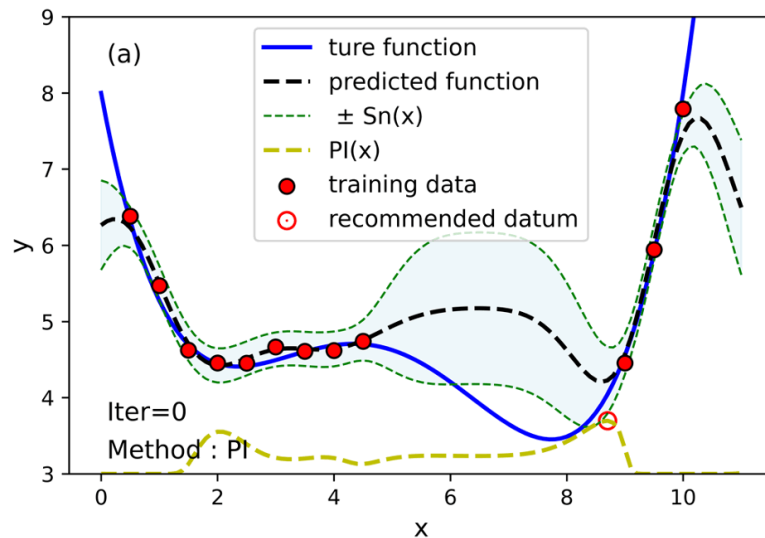
Response surface (响应曲面)

The first and most important step is to **establish the response surface** with collect data pairs

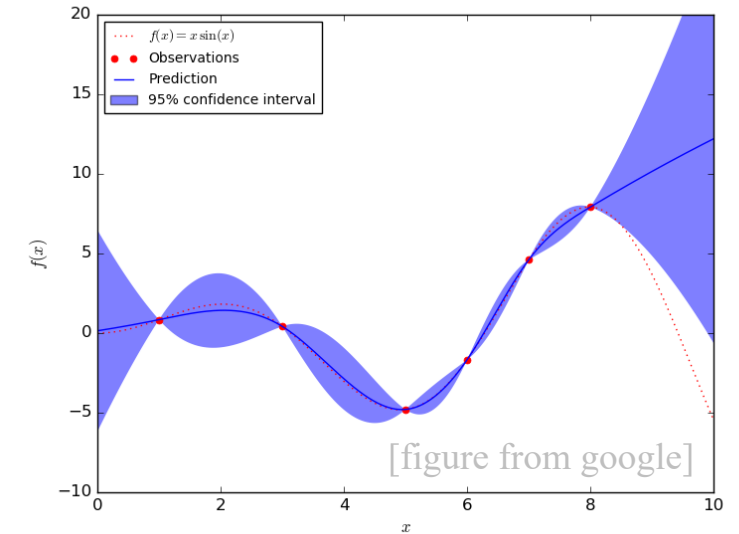


$$\begin{aligned}x_1 &\rightarrow f(x_1) \\x_2 &\rightarrow f(x_2) \\&\dots \\x_n &\rightarrow f(x_n)\end{aligned}$$

→
A ML model



Iterations	x_i	$y_i(x_i)$
1-th	8.7000	4.0418



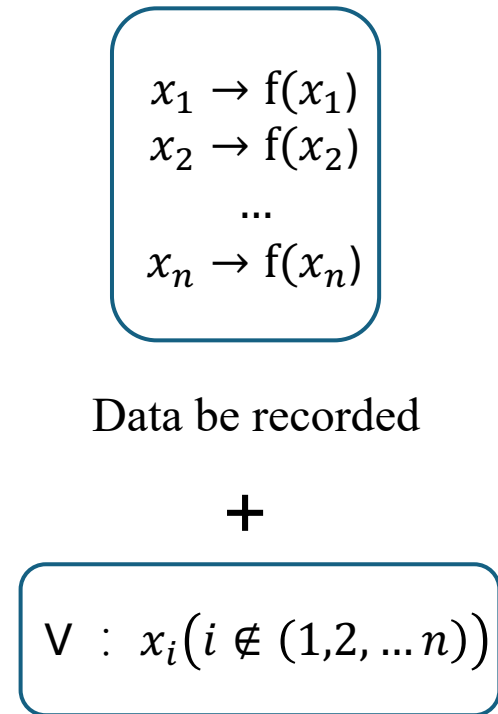
Bgolearn provided **three modules** for establishing the surface :

1. Gaussian Process-Based
2. Boosting Sampling-Based (SVM, RF, MLP, etc.)
3. Multi-model-Based



Virtual/Searching space (搜索空间)

In BGO, the **virtual space** refers to the area constituted by all **unevaluated data** at the current step



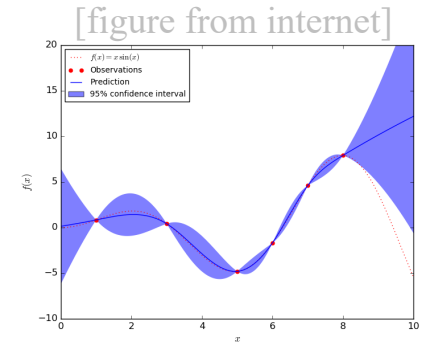
The data within the virtual space, denoted as V

$$\begin{aligned} x_1 &\rightarrow f(x_1) \\ x_2 &\rightarrow f(x_2) \\ &\dots \\ x_n &\rightarrow f(x_n) \end{aligned}$$

ML model

In BGO, the model is not entirely convinced, **highlighting the exploration-exploitation tradeoff**

All data within the virtual space are assessed by the response surface, **taking into account associated uncertainties**



$$V : x_i (i \notin (1, 2, \dots, n))$$

$$G : g(x_i) (i \notin (1, 2, \dots, n))$$

评估候选点的性能



Ranking (效用函数)

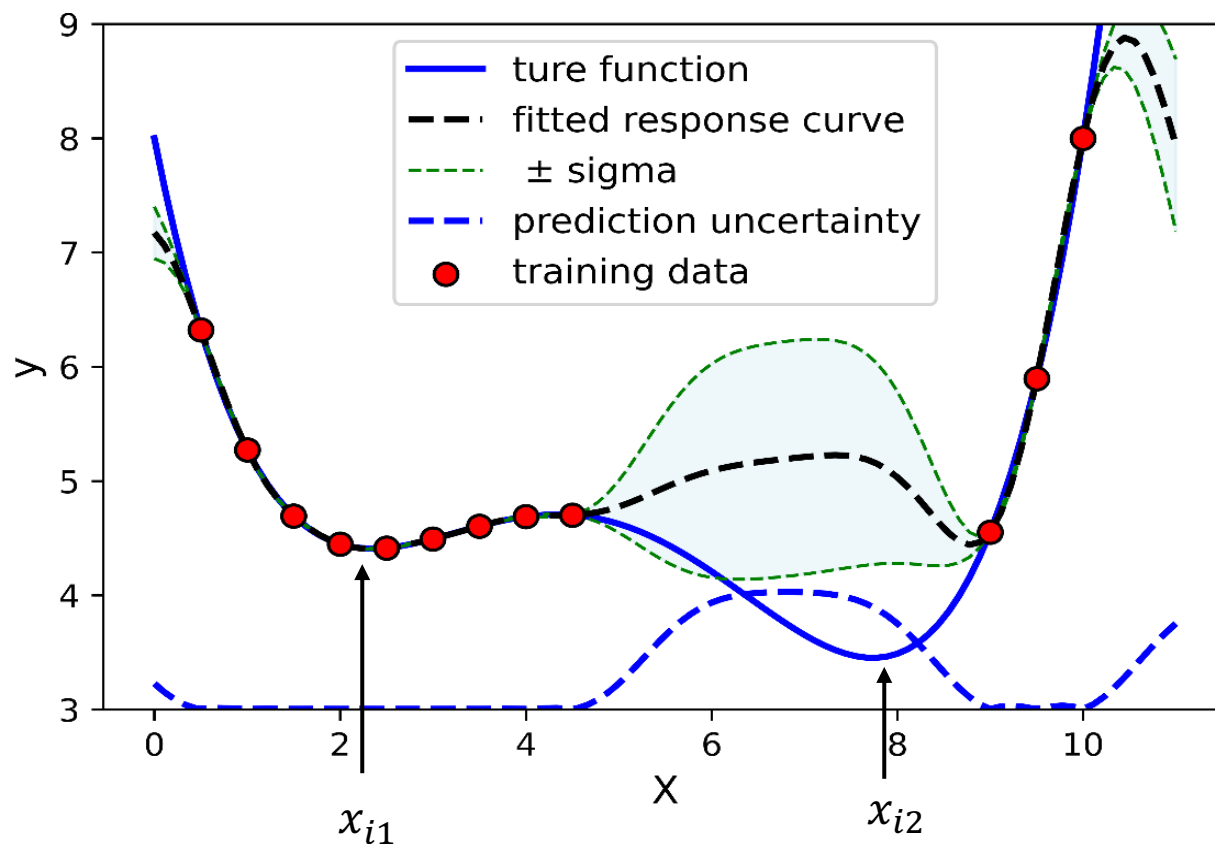
Through the computation of the response surface, we can obtain all the evaluated function values and **their associated uncertainties** for each data point in V

$V : x_i (i \notin (1, 2, \dots, n))$

$G : g(x_i) (i \notin (1, 2, \dots, n))$

$S : \text{std}(x_i) (i \notin (1, 2, \dots, n))$

We use 'g' to denote the evaluated **means** and 'std' to represent the evaluated **standard variance**.



While x_{i1} exhibits the lowest $g(x_{i1})$, however x_{i2} is considered optimal from a global perspective.

How can BGO be guided to recommend solutions closer to x_{i2} ?



Exploration and Exploitation (探索策略)

Pre-defined searching space

How **China** discovered new oil fields ?



[figure from google]

In regions with existing oil fields (**A sites**), the likelihood of discovering additional oil is relatively high. In contrast, previously unexplored regions (**B sites**) offer the potential for discovering larger, untapped reserves.

Prioritizing A sites reduces uncertainty but also limits the chance of major discoveries. Focusing on B sites involves greater uncertainty and may appear less promising, yet it carries the possibility of finding substantially larger reserves. The key challenge is choosing the optimal balance between these two strategies.



风浪越大，鱼越贵

[figure from google]



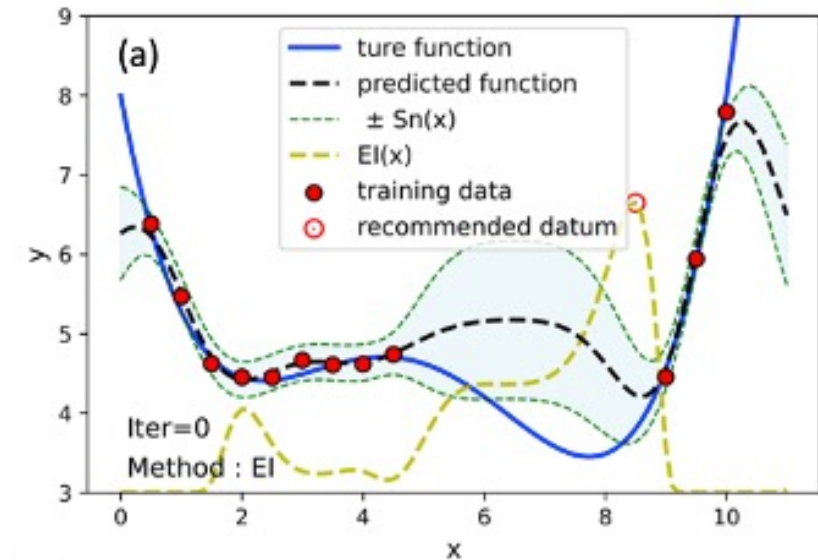
Exploration and Exploitation (探索策略)

The **utility function** is incorporated into BGO, exemplified by the Expectation Improvement algorithm

$$I_n(\mathbf{x}) = \max(f_{min}^{(n)} - g(\mathbf{x}), 0)$$

$$EI_n(\mathbf{x}) = E_{\hat{y}_n(\mathbf{x})} \left(\max(f_{min}^{(n)} - \hat{g}_n(\mathbf{x}), 0) \right)$$

$$EI_n(\mathbf{x}) = s_n \left[\left(\frac{f_{min}^{(n)} - g_n}{s_n} \right) \Phi \left(\frac{f_{min}^{(n)} - g_n}{s_n} \right) + \phi \left(\frac{f_{min}^{(n)} - g_n}{s_n} \right) \right]$$

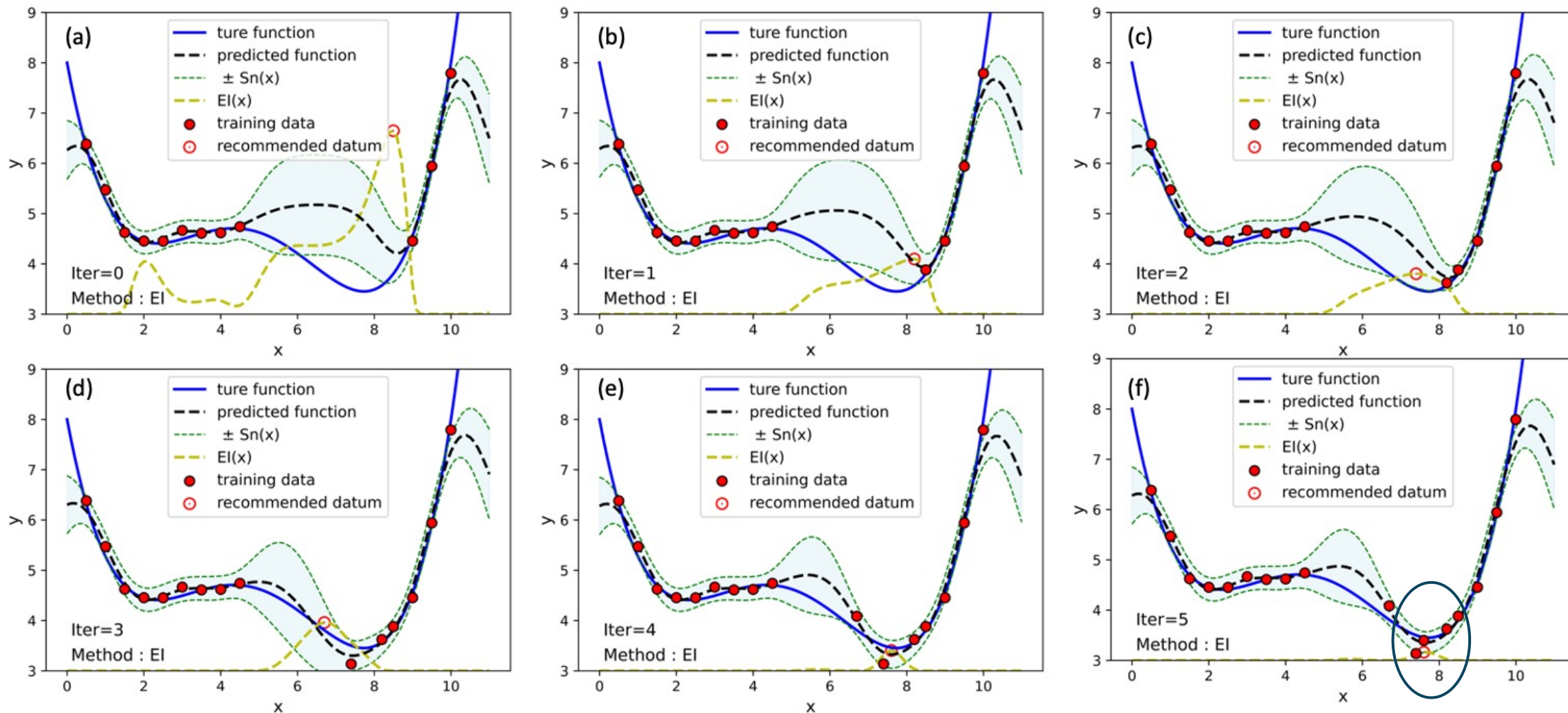


Recommendations are made **by considering both the evaluated values and their associated uncertainties**. The recommended data must then undergo measurement to acquire the real function value (experiments or simulations)



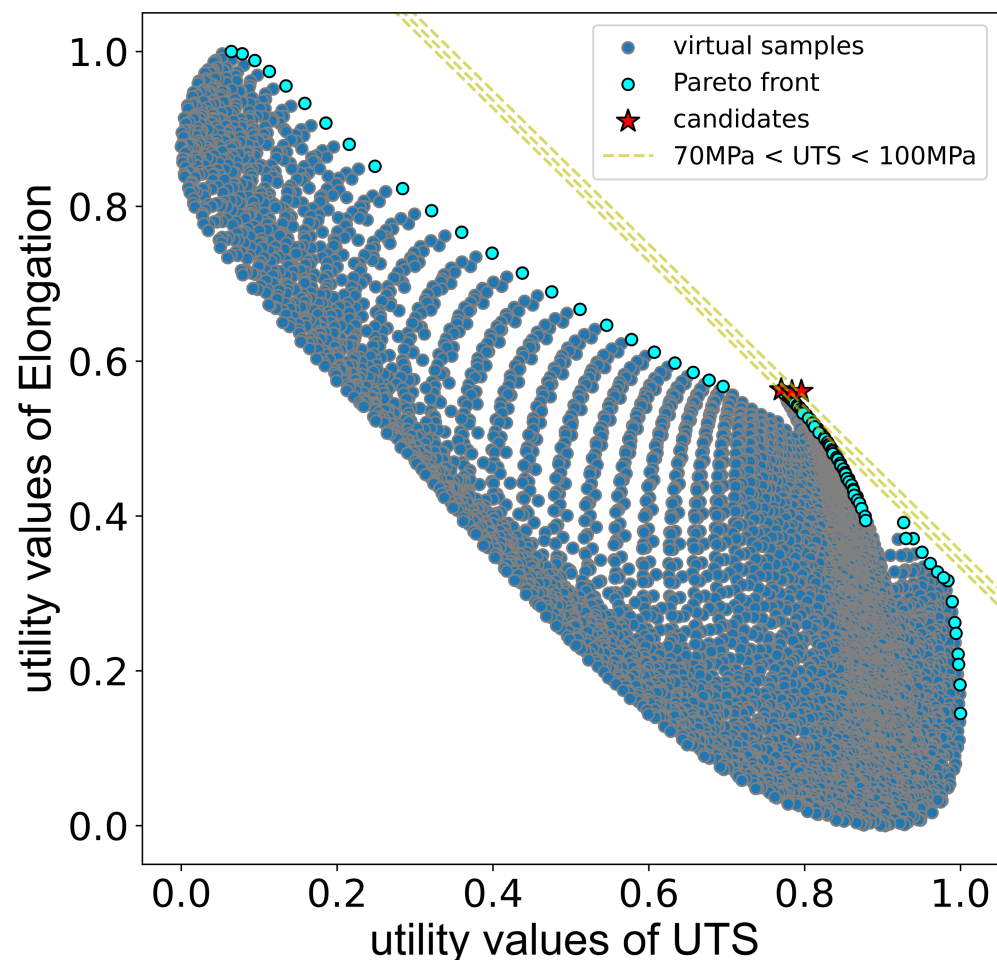
Iteration (迭代优化策略)

The response surface undergoes updates with each iteration





Multi-targets selection : (Mute slide)



In the context of multi-tasking, a straightforward approach involves evaluating each data point in the virtual space twice. This yields the 'score' for each data, and a **Pareto space** is then formed.

Selecting a data point from the Pareto front involves various methods:

1. Diagonal search
2. Distance search
3. Equal value search
4. Crowding distance search
5. ...

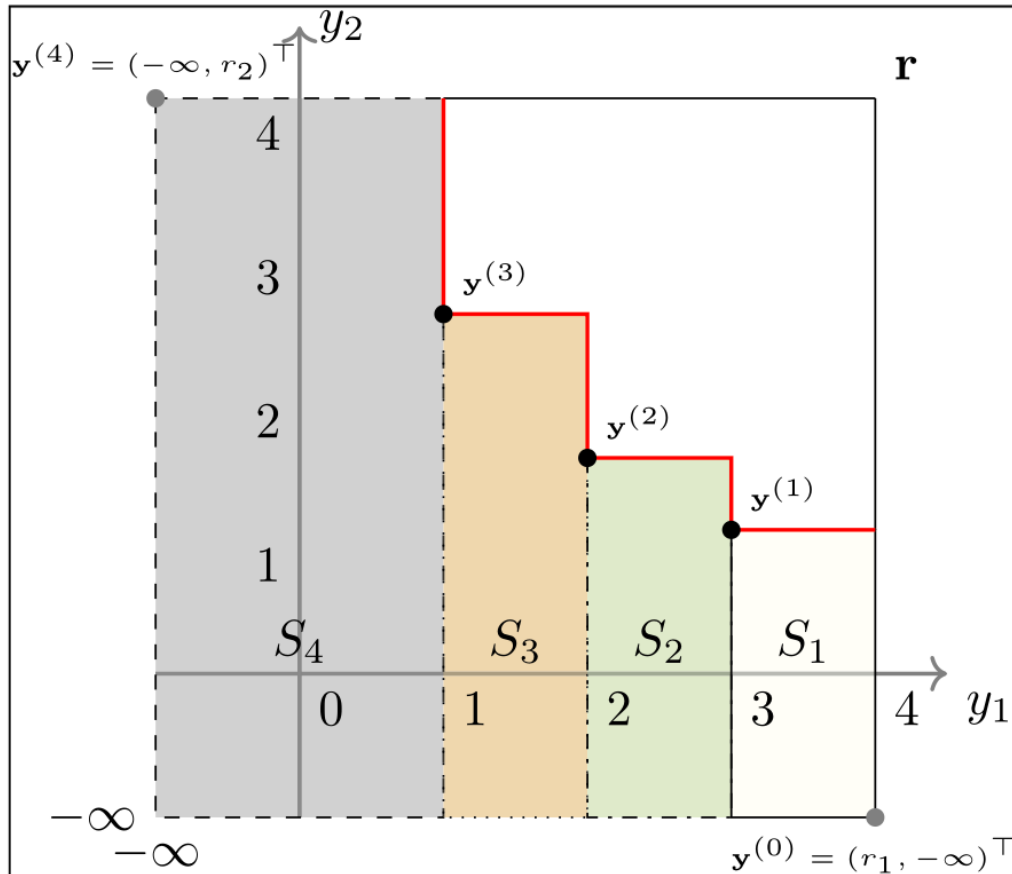
Each method **recommends points located on the front**, contributing to the multi-task optimization process.



Multi-targets BGO : (Mute slide)

In multi-target BGO, the evaluation of infill data focuses on enhancing the influence **across all targets**.

Hypervolume Improvement



Algorithm 1 MOBGO algorithm.

Input: Objective functions $\mathbf{y}$, initialization size μ , termination criterion T_c

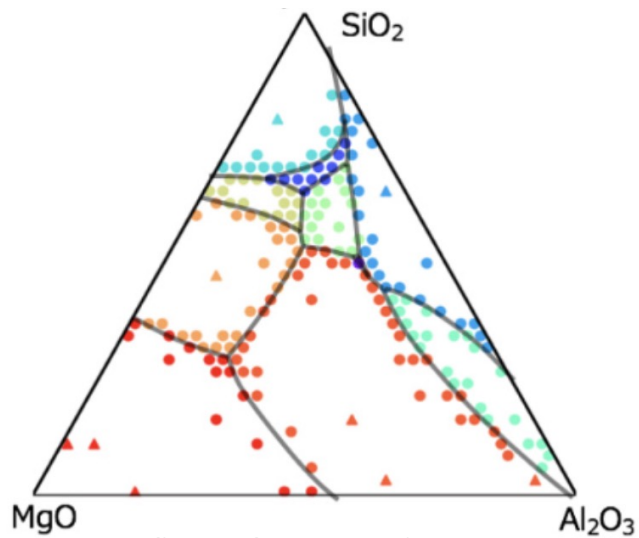
Output: Pareto-front approximation $\mathcal{P}$

- 1 Initialize μ points $(\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(\mu)})$;
- 2 Evaluate the initial set of μ points: $(\mathbf{y}^{(1)} = \mathbf{y}(\mathbf{x}^{(1)}), \dots, \mathbf{y}^{(\mu)} = \mathbf{y}(\mathbf{x}^{(\mu)}))$;
- 3 Store $(\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(\mu)})$ and $(\mathbf{y}^{(1)} = \mathbf{y}(\mathbf{x}^{(1)}), \dots, \mathbf{y}^{(\mu)} = \mathbf{y}(\mathbf{x}^{(\mu)}))$ in D :
 $D = ((\mathbf{x}^{(1)}, \mathbf{y}^{(1)}), \dots, (\mathbf{x}^{(\mu)}, \mathbf{y}^{(\mu)}))$;
- 4 Compute the non-dominated subset of D and store it in $\mathcal{P}$;
- 5 $g = \mu$;
- 6 **while** $g \leq T_c$ **do**
 - 7 Train Kriging models $M_1, \dots, M_d$ based on D ;
 - 8 Use an optimizer (*opt*) to find the promising point $\mathbf{x}^*$ based on surrogate models M , with the infill criterion C ;
 - 9 Update D : $D = D \cup (\mathbf{x}^*, \mathbf{y}(\mathbf{x}^*))$;
 - 10 Update $\mathcal{P}$ as the non-dominated subset of D ;
 - 11 $g = g + 1$;
- 12 Return $\mathcal{P}$



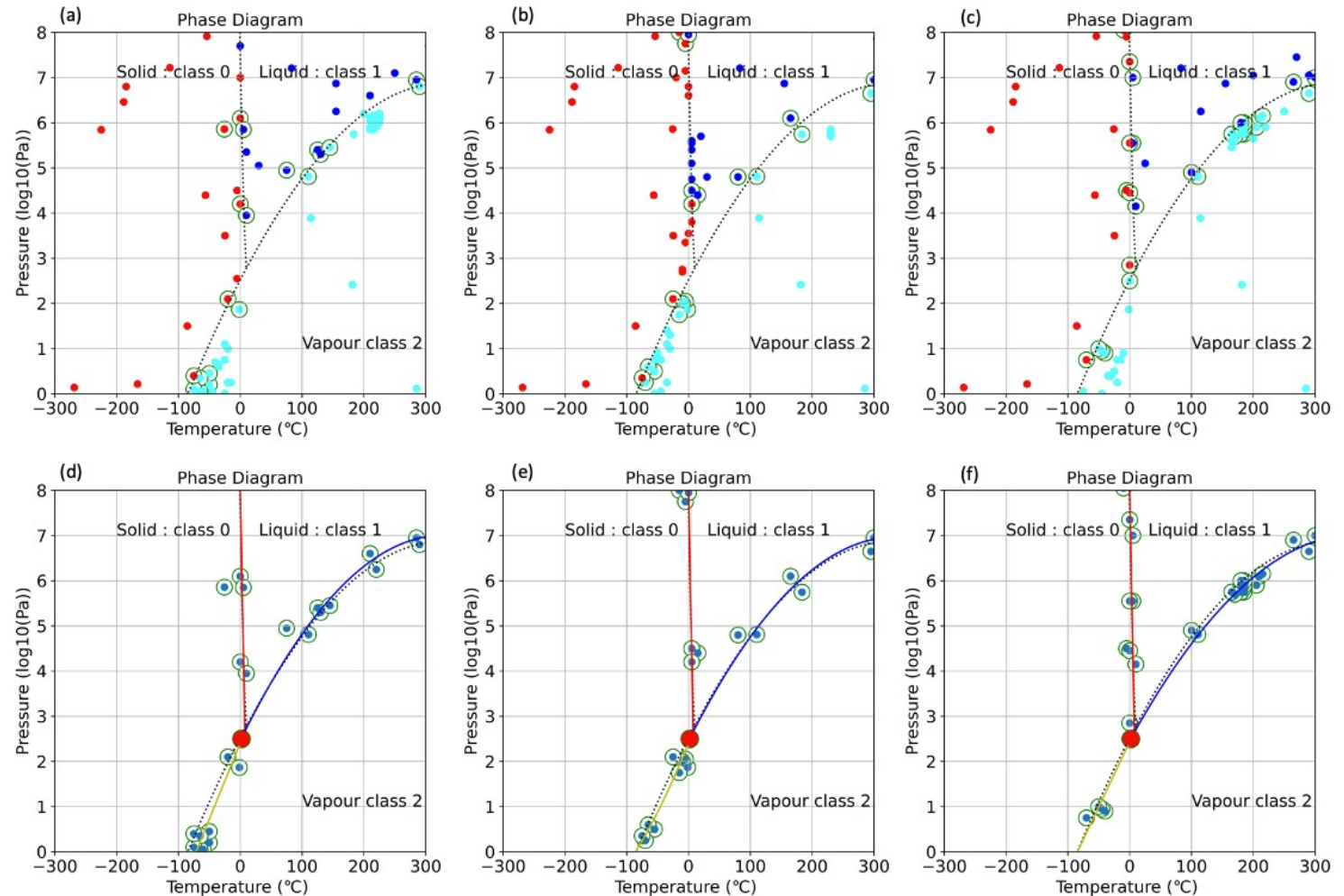
BGO classification (Mute slide)

For **classification** problems, BGO proves to be an excellent optimization method in identifying boundaries between different categories



[figure from google]

Find the phase boundaries





Bgolearn (专为材料设计的优化平台)

Bgolearn is a computational Package that enables material composition-oriented design and performance-oriented optimization. By leveraging existing experimental data, Bgolearn searches for the optimal material composition design within a specified composition space in order to maximize or minimize the desired performance metric.

Only takes three steps

Sstep 1 : `pip install Bgolearn` 

Step 2 :

```
# 导入研究的数据集(已经表征过的样本)
data = pd.read_csv('data.csv')
x = data.iloc[:, :-1]
y = data.iloc[:, -1]
# 设计的虚拟样本, 与x具有相同的维度
vs = pd.read_csv('VS.csv')
```

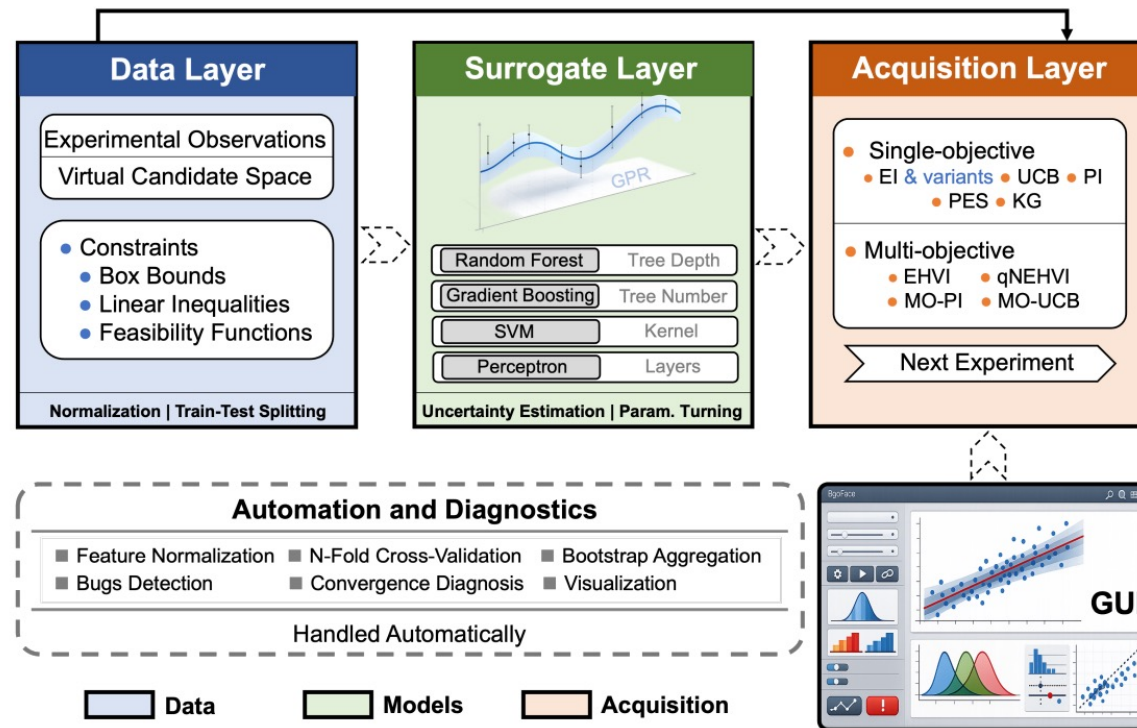
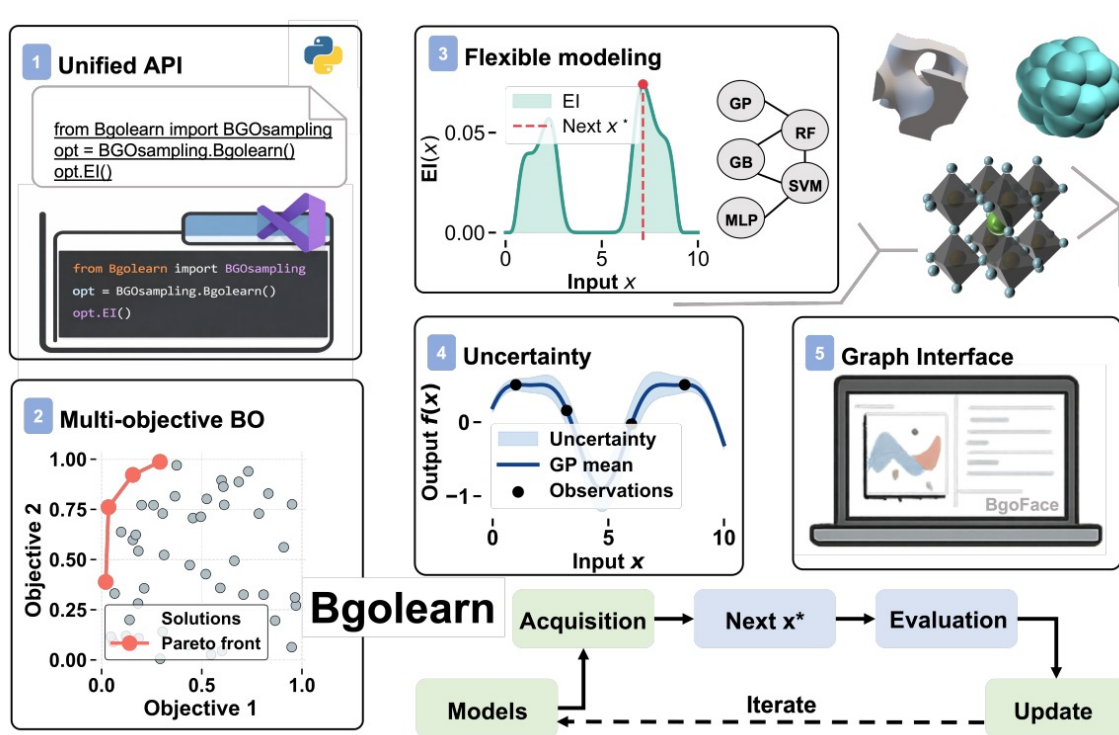
Step 3 :

```
# 实例化类
Bgolearn Bgolearn = BGOS.Bgolearn()
# 传入参数
Mymodel = Bgolearn.fit(data_matrix = x,
Measured_response = y, virtual_samples = vs)
# 通过EI导出结果
Mymodel.EI()
```



Bgolearn (专为材料设计的优化平台)

The Bgolearn package has been downloaded more than **110,000 times** worldwide and has been successfully applied across a broad spectrum of materials discovery problems, spanning metals, functional materials, catalysis, metamaterials, and self-driving laboratories.



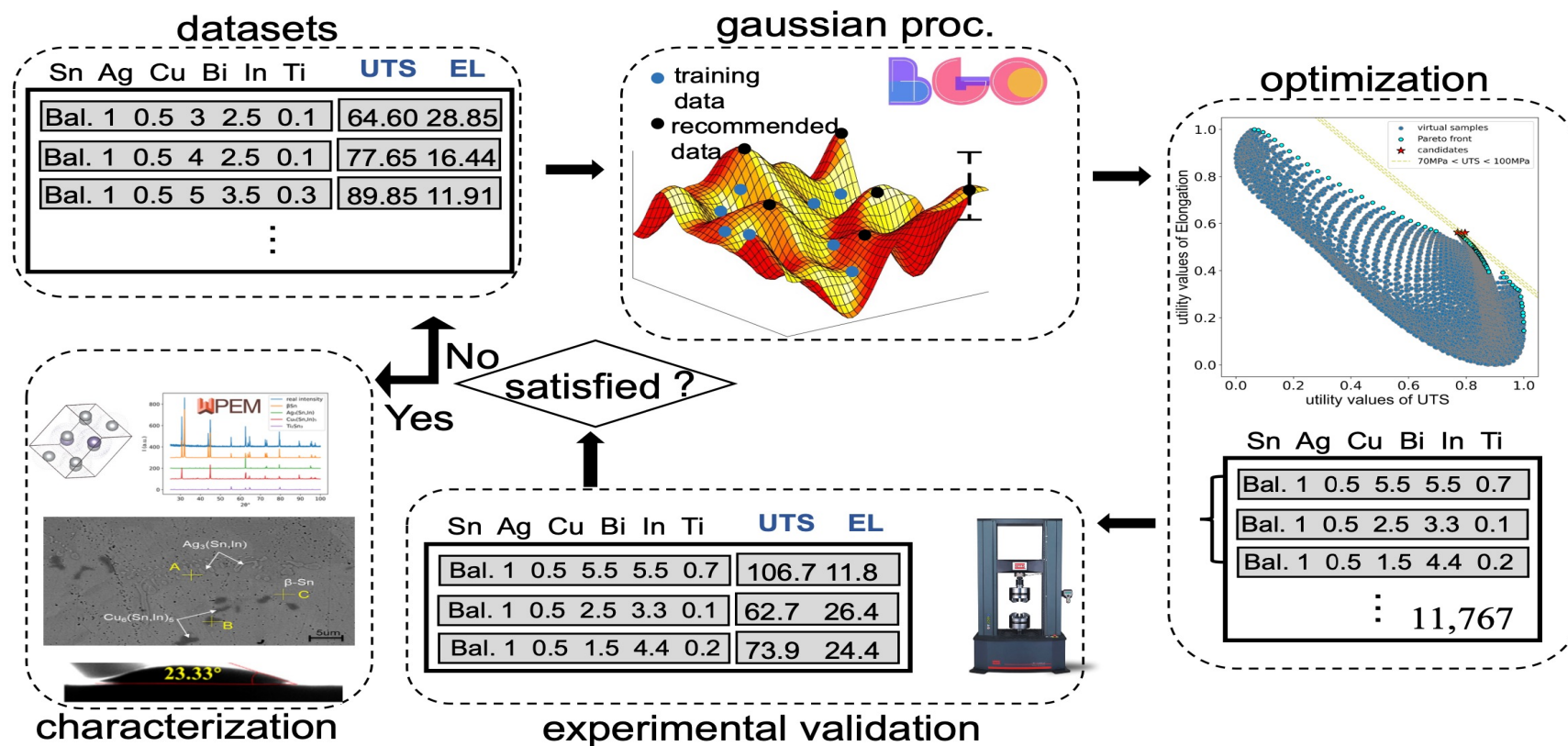
Cao, B et al. Bgolearn: a Unified Bayesian Optimization Framework for Accelerating Materials Discovery. arXiv preprint arXiv:2601.06820.



应用实例

无铅焊料合金优化

Cao 等人利用 Bgolearn 对 Sn – Ag – Cu 焊料合金的多维成分空间进行探索，重点解决强度与延展性之间的权衡问题。通过高斯过程回归结合 UCB 采集函数，Bgolearn 在仅三轮实验内便识别出一种低银含量合金，实现了优异的力学性能平衡。

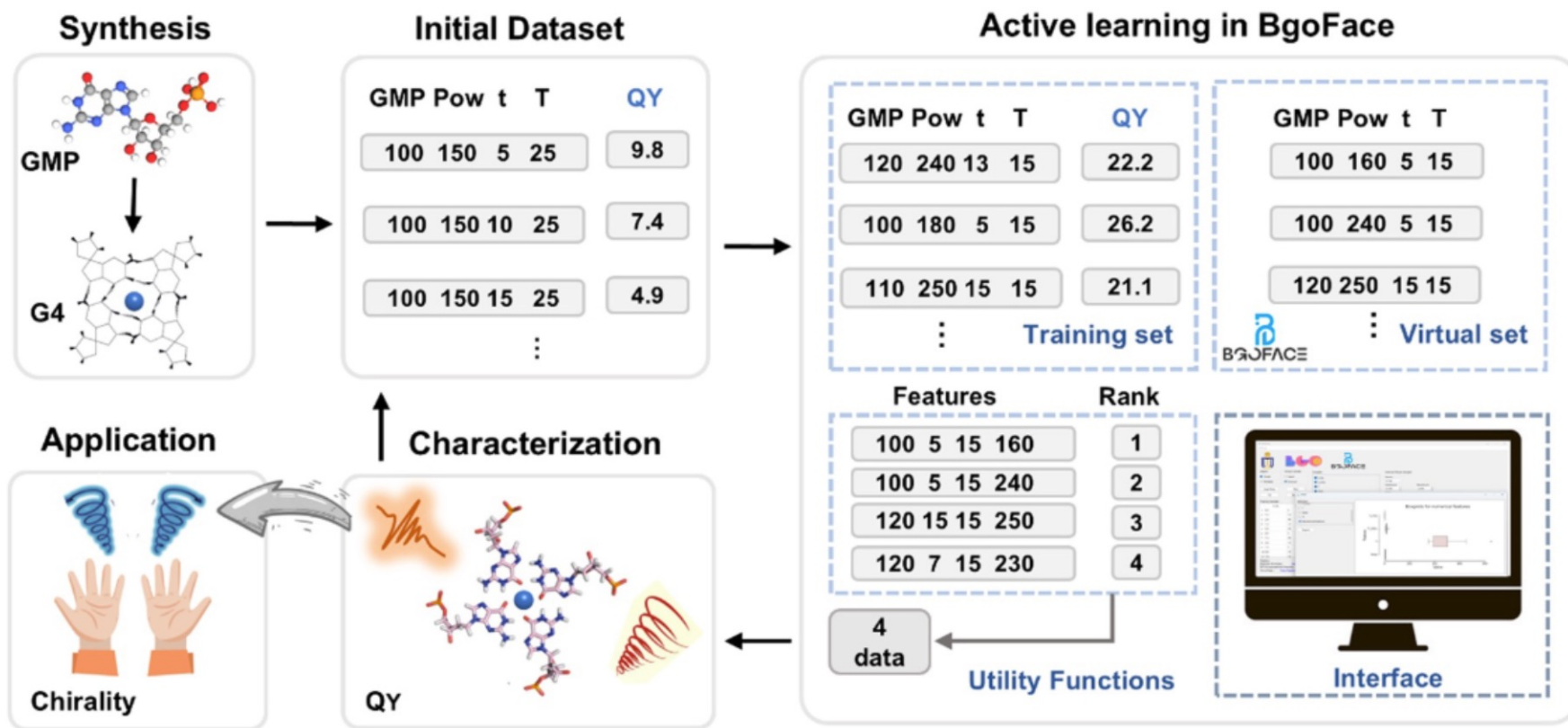




应用实例

纳米酶与发光材料设计

Li 等人利用 Bgolearn 设计高量子产率纳米酶体系。基于少量 G-四链体圆偏振发光材料数据，通过多轮主动学习迭代，显著缩小实验空间，最终发现量子产率几乎提升一倍的材料，同时大幅减少高成本合成实验。



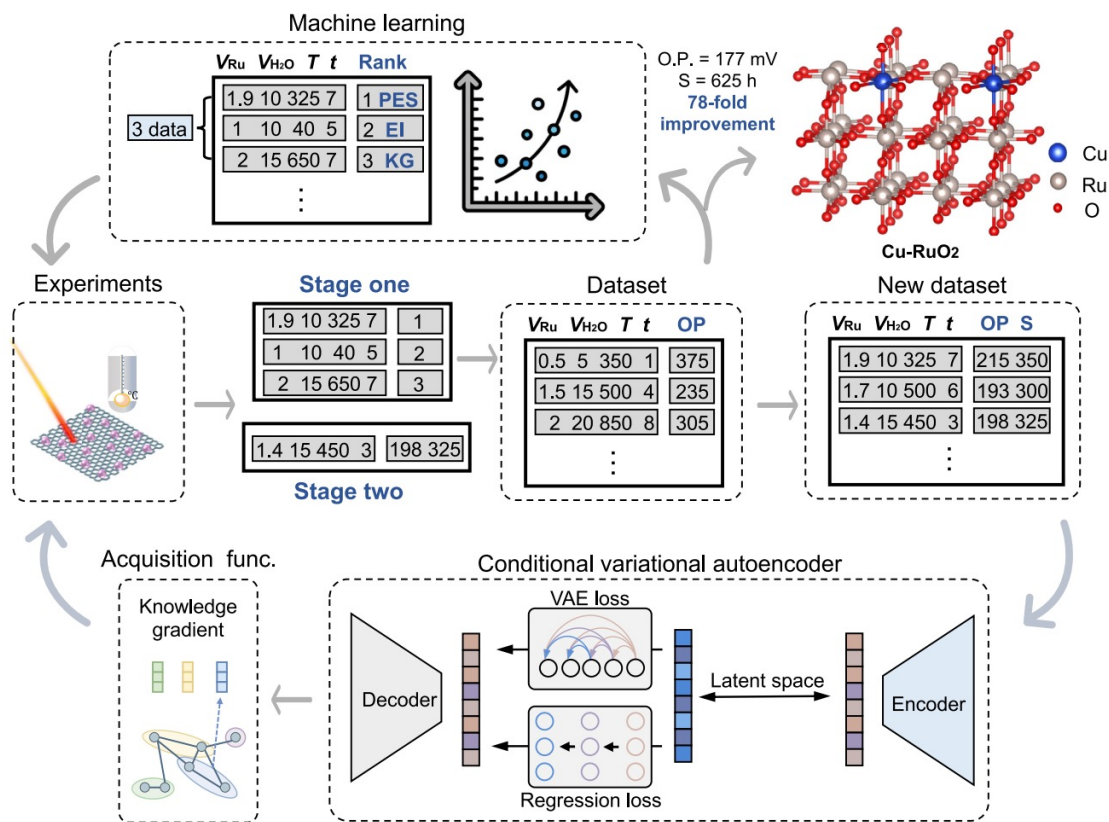


应用实例

酸性 OER 催化剂材料设计

Cao 等人提出一种两阶段优化策略：

第一阶段使用 Bgolearn 最大化 OER 活性，第二阶段在高活性子空间中进一步进行稳定性自适应优化。该策略有效降低稳定性测试成本，并成功发现兼具高活性与高耐久性的酸性 OER 催化剂。

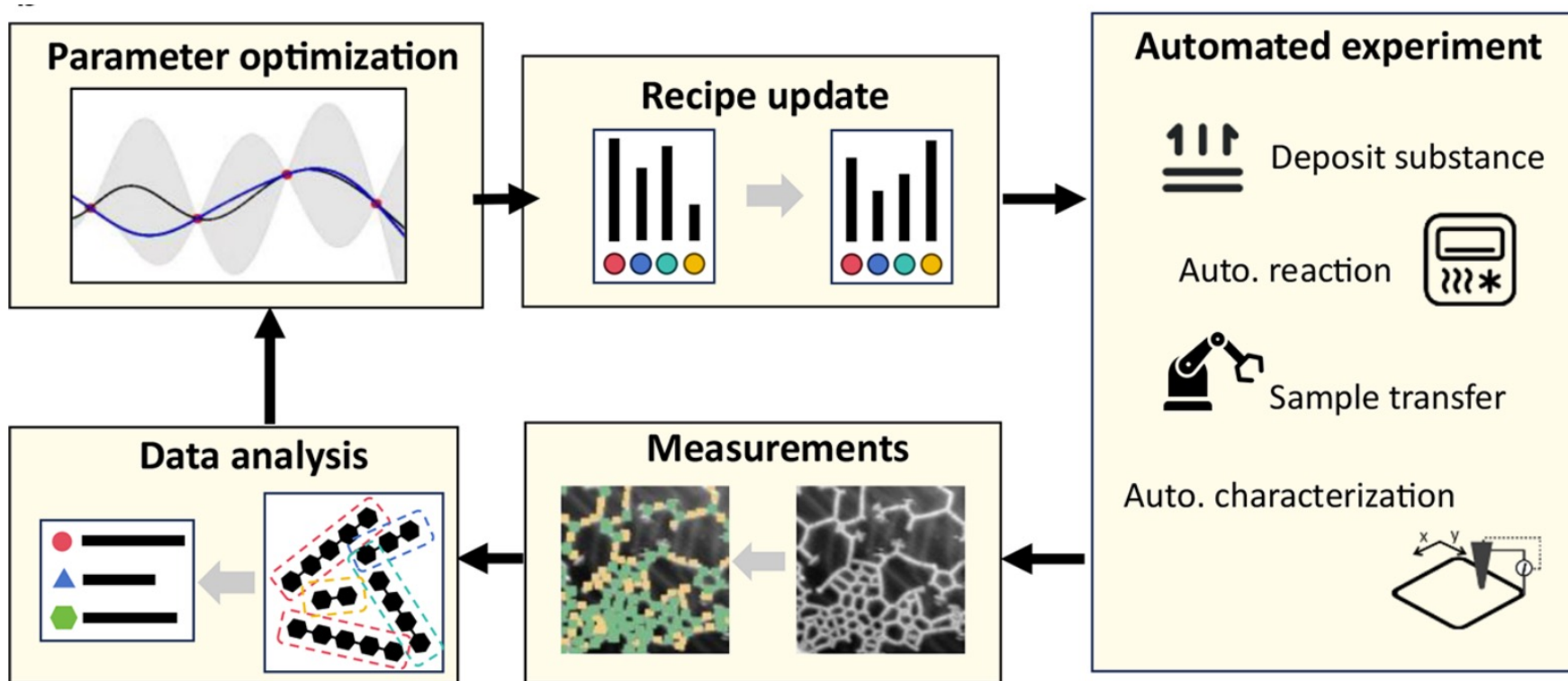




应用实例

自主表面合成实验平台

Li 等人将 Bgolearn 集成到超高真空条件下的自主表面合成实验平台中，用于石墨烯纳米带的快速优化合成。该平台结合机器人控制、原位扫描隧道显微镜与贝叶斯优化，由 Bgolearn 作为核心决策引擎，构建合成参数与材料结构之间的概率模型，并迭代选择最具信息量的实验。验证了 Bgolearn 在闭环、自主材料发现中的能力。





代码展示

主要资源（展示CodeDemo ~15mins）

<https://bgolearn.netlify.app/> （说明书-英文）

<https://bgolearn-chi.netlify.app/> （说明书-中文）

<https://github.com/Bgolearn/BgoFace/releases> （软件界面）

<https://github.com/Bin-Cao/Bgolearn> （代码地址、项目主页）

https://mp.weixin.qq.com/s/S_5An5scrqcxbmvINU5JQ （微信推送）

https://www.bilibili.com/video/BV1LTtLeaEZp/?spm_id_from=333.337.search-card.all.click （bilibili 视频）



Thank you!

If you use the PPT, please quote it as follows :

Cao, B et al. Bgolearn: a Unified Bayesian Optimization Framework for Accelerating Materials Discovery. arXiv preprint arXiv:2601.06820.